Tsao-Lun Chen

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Professional Summary

Interdisciplinary AI researcher with electrical engineering training from National Taiwan University of Science and Technology (NTUST) and ongoing graduate study in oral biology at National Taiwan University (NTU). My AI research focuses on robust semi-supervised learning, dental AI, and multimodal agent system development; my current lab work focuses on stem cells and extracellular vesicles, with the goal of applying AI to biomedical and regenerative medicine research.

Selected Experience

Intelligent Manufacturing Innovation Center (IMIC), NTUST — Taipei, Taiwan
Research Assistant — Jul. 2023 – Feb. 2026

- Wrote the NVIDIA Academic Grant proposal on semi-supervised medical image segmentation under limited-annotation settings; the project received 21K A100 GPU-hours and later evolved into **USE**, a first-author study on unlabeled data quality control for robust semi-supervised learning.
- Participated in the NTUST × NVIDIA DGX Spark initiative, contributing to procurement, implementation, interview preparation, article/copywriting, and recording for local AI workflow demonstrations.
- Managed shared lab compute resources and supported research execution across student projects, including experiment setup, server usage coordination, and model development workflows.

Robust SSL Evaluation Project — Taipei, Taiwan
First-Author Research, C-Score for Open-World SSL Robustness Assessment — Jan. 2026 – Apr. 2026

- Conducted a follow-up study to **USE** on the evaluation of pseudo-label-based semi-supervised learning under open-world unlabeled contamination, motivated by the limitation of accuracy-only assessment.
- Proposed **C-Score**, a compact diagnostic framework spanning prediction, feature, and optimization spaces through Pseudo-label Entropy (PLE), Class Concentration Index (CCI), Semantic Drift (Sem-Drift), and Gradient Alignment (Grad-Align).
- Showed that clean test accuracy alone can mask degradation under open-world unlabeled contamination, while **CCI** provided the clearest and most consistent signal of hidden collapse across contamination settings.

NTU Graduate Institute of Oral Biology — Taipei, Taiwan
Research Collaboration, AI for Dental Panoramic Radiographs — Jun. 2023 – Feb. 2025

- Served as the engineering contributor and second author on a multinational dental AI study spanning 6,669 dental panoramic radiographs from the Netherlands, Brazil, and Taiwan.
- Built model development pipelines covering detection, segmentation, post-processing, and visualization for per-tooth dental finding assessment.
- Contributed to a system that achieved **96.2% macro AUC** across 8 findings, showed robust cross-site generalization, and processed images substantially faster than human readers.

NTU CSIE × NTU Graduate Institute of Oral Biology — Taipei, Taiwan
Ongoing Collaboration, Multimodal Caries Diagnosis on Panoramic Radiographs — Ongoing

- Extended prior panoramic radiograph research toward caries diagnosis, a clinically difficult task on panoramic radiographs, through multimodal / VLM-based modeling.
- Explored image-text fusion strategies to incorporate dental knowledge and improve semantic grounding for caries assessment in panoramic imaging.

Temporal Knowledge Graph Reasoning Project — Taipei, Taiwan
Research Collaboration, Reachability-Aware Pretraining for TKG Reasoning — Oct. 2025 – Mar. 2026

- Co-developed **RAPTOR**, a self-supervised pretraining method for RL-based temporal knowledge graph reasoning under sparse rewards and large time-dependent action spaces.
- Focused on reachability-aware pretraining to reduce wasted exploration and improve training efficiency for target-oriented path reasoning on temporal graphs.

National Taiwan University of Science and Technology Taipei, Taiwan
M.S. Thesis Researcher, Generalized Zero-Shot Learning Sep. 2020 – Aug. 2022

- Conducted M.S. thesis research under Prof. Shun-Feng Su on generalized zero-shot learning with semantic-visual alignment.
- Proposed an attention fusion module with GAN-based feature generation for end-to-end GZSL, linking multimodal attention with synthesized unseen-class features.
- Achieved **75.8%** on CUB, **72.3%** on AWA2, and **42.5%** on SUN, establishing an early foundation for later work on multimodal learning and robustness.

Selected Publications and Preprints

- **Tsao-Lun Chen**, Chien-Liang Liu, Tzu-Ming Harry Hsu, Tai-Hsien Wu, Chi-Cheng Fu, Han-Yi E. Chou, and Shun-Feng Su. *USE: Uncertainty Structure Estimation for Robust Semi-Supervised Learning*. arXiv preprint, 2026.
- **Tsao-Lun Chen**, Chi-Cheng Fu, Han-Yi E. Chou, and Shun-Feng Su. *C-Score: Beyond Accuracy for Robustness Assessment in Semi-Supervised Learning under Open-World Unlabeled Contamination*. Accepted for oral presentation at *International Conference on System Science and Engineering (ICSSE)*, 2026.
- Yin-Chih Chelsea Wang, **Tsao-Lun Chen**, et al. *Artificial Intelligence to Assess Dental Findings from Panoramic Radiographs – A Multinational Study*. arXiv preprint, 2025. Submitted to *Dentomaxillofacial Radiology*.
- Chien-Liang Liu, **Tsao-Lun Chen**. *Reachability-Aware Pretraining for Efficient Target-Oriented Path Exploration in Temporal Knowledge Graph Reasoning*. Submitted to ACL Rolling Review, 2026.
- **Tsao-Lun Chen**, Wei-Tang Tseng. *Automatic Reference Current Architecture in Computing in Memory by MRAM*. ECICE, 2019.

Education

National Taiwan University Taipei, Taiwan
M.S. in Graduate Institute of Oral Biology Feb. 2026 – Present
Advisor: Han-Yi E. Chou

National Taiwan University of Science and Technology Taipei, Taiwan
M.S. in Electrical Engineering (GPA: 3.99/4.3) Sep. 2020 – Jun. 2022
Advisor: Shun-Feng Su